

Pseudo-terminal will not be allocated because stdin is not a terminal

Asked 12 years ago Modified 9 months ago Viewed 676k times



472



I am trying to write a shell script that creates some directories on a remote server and then uses scp to copy files from my local machine onto the remote. Here's what I have so far:

```
ssh -t user@server<<EOT
DEP_ROOT='/home/matthewr/releases'
datestamp=$(date +%Y%m%d%H%M%S)
REL_DIR=$DEP_ROOT"/"$datestamp
if [ ! -d "$DEP_ROOT" ]; then
    echo "creating the root directory"
    mkdir $DEP_ROOT
fi
mkdir $REL_DIR
exit
EOT

scp ./dir1 user@server:$REL_DIR
scp ./dir2 user@server:$REL_DIR
```

Whenever I run it I get this message:

Pseudo-terminal will not be allocated because stdin is not a terminal.

And the script just hangs forever.

My public key is trusted on the server and I can run all the commands outside of the script just fine. Any ideas?

linux

bash

shell

ssh

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edited Dec 14, 2015 at 15:27



Claudio

Claud 10.6k ● 4 ● 31 ● 71

asked Aug 18, 2011 at 22:38



Matthew

12.9k ● 6 ● 42 ● 45

4 You can simply specify the terminal to use like `ssh user@server /bin/bash <<EOT...` – Buzut Apr 11, 2016 at 1:07

3 @Buzut: You probably mean *shell*, but, yes, specifying `/bin/bash` explicitly is one way to avoid the problem. – mklement0 Jan 22, 2017 at 20:17

1 @mklement0 indeed, that's what I meant. Thx for correcting that ;) – Buzut Jan 22, 2017 at 21:52

In case anyone has the same problem I have, it seems that running `ssh` from `subprocess.run("ssh ...")` in `python` is different when it is running under `pytest`. The later does something funny with `stdin` while the former doesn't. So your code works, but the tests fail. Adding `-tt` to the command line fixed it. – [Mark Lakata](#) May 18, 2022 at 21:26

9 Answers

Sorted by:

Highest score (default) 

Try `ssh -t -t` (or `ssh -tt` for short) to force pseudo-tty allocation even if stdin isn't a terminal.

647

See also: [Terminating SSH session executed by bash script](#)



From ssh manpage:



`-T` Disable pseudo-tty allocation.

`-t` Force pseudo-tty allocation. This can be used to execute arbitrary screen-based programs on a remote machine, **which** can be very useful, e.g. when implementing menu services. Multiple `-t` options force **tty** allocation, even **if** ssh has no **local tty**.

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edited Dec 2, 2017 at 14:30

answered Aug 19, 2011 at 13:14

[Willem van Ketwisch](#)

5,676 ● 7 ● 49 ● 57

[carok](#)

6,542 ● 1 ● 14 ● 2

- 1 This suggestion fixed my problem. I have a script calling another one in the background: "source myscript.sh &". Inside "myscript.sh" there is ssh that is using "sudo" like this: ssh user@host "sudo <command>". Using a single `-t` parameter didn't help, only "`-t -t`". – [OutputLogic](#) Apr 17, 2012 at 0:18 
- 23 I am having a similar issue in a script that is run here. I added the `-t -t` but now I am getting a new error. "tcgetattr: Inappropriate ioctl for device" – [JLZenor](#) Dec 18, 2012 at 19:32
- 6 What does it mean when it says "even if ssh has no local tty"? – [CMCDragonkai](#) Apr 29, 2016 at 7:42
- 2 @CMCDragonkai: If you provide commands to `ssh` via *stdin*, its *stdin*, due to the input redirection, is no longer connected to a *terminal*, so in that sense "ssh has no local tty" anymore. It is in this scenario that a *single* `-t` is insufficient to allocate a pty and `-t -t` (`-tt`) must be used instead. That said, you can avoid all this if you don't provide commands via *stdin* to begin with, and use a command-line *argument* instead. – [mklement0](#) Jan 22, 2017 at 16:25
- 5 So `-t` and `-t -t` all about allocating a pty on the remote side of the connection. And the problem is that local side *stdin* is not a pty, so you can either fully disable it with `-T`, or force a remote side pty. The choice depends on whether the commands you are running on remote side relies on there being a tty or not. But it is puzzling as to why ssh is hanging as is claimed by the question. – [CMCDragonkai](#) Jan 23, 2017 at 7:22 



Also with option `-T` from [manual](#)